Phase 1 Feedback and Next Phase Work

I have gone through your Phase 1 report and the GitHub repository. The work has improved a lot. Early stopping works in every run, three seeds are used, the input size is chosen on validation, and the split is fixed. Please fix the points in Section 1, then do the next phase in Section 2. You do not need my approval between the steps. The checklists will tell you whether a step is done. Send me everything together at the end.

1. Corrections to Phase 1

1. Do not write that 224 x 224 is better than 96 x 96. The difference is small and within the variation between seeds. Write that they are comparable and that 96 was chosen because it is about three times faster.
2. Scratch GroupNorm was trained with the same learning rate as BatchNorm. Train it with learning rates 0.0003, 0.001 and 0.003 (three seeds each), pick the best on validation, and use that from now on. Write the BatchNorm and GroupNorm comparison as "under our training setup".
3. Pretrained and scratch models used different learning rates. Say this in the report and do not credit the whole gain to pretraining.
4. Repository:
   - State which folder and which commit gave the report numbers.
   - Replace requirements.txt with exact versions (pip freeze).
   - Upload the best_model.pt files as a GitHub release or a shared archive, with a list of file names and SHA-256 hashes.
   - Rewrite the README for the Phase 1 settings and add the commands that reproduce the results.
   - Tag the final commit as phase1-final.

Checklist: every number in the report comes from one tagged commit, and anyone can install the same versions and download the models.

2. Next phase work

Use the pretrained ResNet-18 with BatchNorm at 96 x 96 for everything below. At the very end, repeat one comparison with the scratch GroupNorm model (see Part C).

Rules for all runs:

- Only the training set is changed. Validation and test sets stay balanced and the same everywhere.
- Choose all settings on validation. Use the test set only once, at the end.
- Stop training when validation loss does not improve for 10 epochs.
- Final results use five seeds (42, 123, 2024, 7, 99). Report mean and standard deviation.
- Write one line per run in results.csv. Make every table and figure from this file, and never type numbers by hand.

Part A: Make the imbalanced data

Dirichlet split (clients with different class mixes):

- Split the training set among 10 clients with alpha = 0.1, 0.5, 1 and 10. Small alpha means very

different clients.

- Each client must get at least 10 images. If not, redo the split with the same seed.
- Use the NIID-Bench code (label skew option) and cite its paper.
- Save each split as a json file.

Long-tailed split (one model, rare classes):

- Keep the largest class complete and reduce the other classes step by step so that the largest class is 10, 50 or 100 times bigger than the smallest. The number of images in class k is n_max times ratio raised to the power (minus k divided by 9), for k = 0 to 9.
- Fix the class order with a seed and never change it.

Checklist: running the code twice with the same seed gives the same files, and the class counts plot looks as expected (very uneven at alpha 0.1 and ratio 100).

Part B: Train with normal cross-entropy

- Dirichlet: train each client's model on its own data and test it on the global test set. Also give the pooled-data model from Phase 1 as the reference.
- Long-tailed: train one model for each ratio.
- Save accuracy, macro-F1 and per-class recall for every run.

Checklist: results get worse as alpha gets smaller and as the ratio gets larger. If the trend jumps up and down, check your code before going on. A model trained on a skewed split will obviously do worse, so this alone is not a result.

Part C: Try better losses

Compare three losses:

- Cross-entropy (the reference).
- Class-balanced loss (Cui et al., CVPR 2019), beta from 0.99, 0.999, 0.9999.
- Focal loss (Lin et al., ICCV 2017), gamma from 1, 2.

Choose beta and gamma on validation macro-F1, fix them, and then run the final five seeds. Run this for both the long-tailed and the Dirichlet setting. Finally, repeat the ratio 100 case once with scratch GroupNorm, comparing cross-entropy with the best loss.

Checklist: beta and gamma were chosen before the test set was used, all numbers have five seeds, and the best loss is compared with cross-entropy in every setting. If a loss does not help somewhere, report it as it is.

3. What goes into the paper

A conference paper has room for three tables and four figures. Everything else goes into a supplementary file or the GitHub repository (for example the per-class recall tables, the chosen beta and gamma values, the GroupNorm learning rate results and the training curves).

Table 1. Baseline results (real numbers from your report)

Model	Input	Test accuracy	Test macro-F1	Train time (min)
ImageNet, BN	64	98.32 ± 0.11	98.27 ± 0.11	8.84

ImageNet, BN	96	98.64 ± 0.09	98.59 ± 0.08	10.84
ImageNet, BN	224	98.94 ± 0.24	98.90 ± 0.25	33.44
Scratch, BN	96	96.87 ± 0.24	96.76 ± 0.26	16.86
Scratch, GN	96	95.31 ± 0.20	95.19 ± 0.19	13.47

Table 2. Long-tailed EuroSAT, test macro-F1 (dummy values, layout only)

Loss	Ratio 10	Ratio 50	Ratio 100
Cross-entropy	95.1 ± 0.4	88.3 ± 0.9	82.7 ± 1.3
Class-balanced	95.6 ± 0.3	90.4 ± 0.7	86.0 ± 1.0
Focal	95.3 ± 0.4	89.1 ± 0.8	84.1 ± 1.2

Table 3. Dirichlet split with 10 clients, global test macro-F1 (dummy values, layout only)

Loss	alpha 0.1	alpha 0.5	alpha 1	alpha 10
Cross-entropy	31.4 ± 2.1	58.2 ± 1.8	69.5 ± 1.5	88.7 ± 0.9
Class-balanced	36.0 ± 2.0	62.1 ± 1.7	72.0 ± 1.4	89.4 ± 0.8
Focal	34.1 ± 2.2	60.3 ± 1.8	70.8 ± 1.5	89.0 ± 0.9

The values in Tables 2 and 3 are dummy numbers only to show the format. Do not copy them anywhere.

Figures (PDF, same colour for the same method everywhere, labelled axes):

1. Client against class heat map, one panel for alpha 0.1 and one for alpha 10.
2. Macro-F1 against ratio and against alpha, one line per loss, with error bars.
3. Per-class recall at ratio 100, cross-entropy against the best loss.
4. Confusion matrix of the best Phase 1 model on the test set.

Under each table and figure write two or three sentences on what it shows and what it does not show.

4. What to write

Write these sections in LaTeX in this order:

1. Related work (EuroSAT and remote sensing, non-IID data, imbalance losses, federated learning in remote sensing). End with what we do that the closest paper does not.
2. Problem definition (task, Dirichlet split, long-tailed split, metrics).
3. Method (model, the two losses with equations).
4. Experimental setup (dataset, split, settings, seeds, hardware and software versions).
5. Results (Tables 1 to 3, Figures 1 to 4).
6. Limitations: random image-level split, one dataset, RGB only, five seeds, and Dirichlet splits only approximate real regional differences.